Generating functional and Green's functions Sept 29

$$D(x',T;x,0) = \int \mathcal{D}x(t)\; e^{\frac{i}{\hbar}\int_0^T dt\, L}$$

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} m \Omega^2 x^2 + J(t)\, x$$

$$x(t) = x_{cl}(t) + y(t) \quad , \quad BC: \; x_{cl}(0) = x, \; x_{cl}(T) = x'.$$

$$x_{cl}: \quad \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0$$

$$\Rightarrow \quad m \ddot{x}_{cl} + m\Omega^2 x_{cl} - J(t) = 0$$

$$\ddot{x}_{cl} + \Omega^2 x_{cl} = J(t)/m$$

solves

$$\Rightarrow \quad x_{cl}(t) = x_{hom}(t) + x_{inhom}(t)$$

solves $\ddot{x}_{hom} + \Omega^2 x_{hom} = 0 \Rightarrow x_{hom}(t) = A e^{i\Omega t} + B e^{-i\Omega t}$

To find $x_{inh}(t)$, we need to introduce <u>Green's function</u> G

$$\left(\frac{d^2}{dt^2} + \Omega^2\right) G(t-t') = -\delta(t-t')$$

Then $$x_{inh}(t) = -\int_0^T dt'\; G(t-t')\, \frac{J(t')}{m}$$

because $$\left(\frac{d^2}{dt^2} + \Omega^2\right) x_{inh}(t) = -\int_0^T dt' \underbrace{\left(\frac{d^2}{dt^2} + \Omega^2\right) G(t-t')}_{-\delta(t-t') \text{ by assumption}} \frac{J(t')}{m}$$

$$= \int_0^T dt'\; \delta(t-t')\, \frac{J(t')}{m} = \frac{J(t)}{m}, \text{ so that}$$

$x_{inh}(t)$ indeed solves inhomogeneous eqn.

How to find $G(t-t')$?

① Fourier transformation

$$G(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} G(\omega)$$

$$\Rightarrow \int \frac{d\omega}{2\pi}\left(-\omega^2 + \Omega^2\right) G(\omega) e^{-i\omega(t-t')} = \underbrace{-\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')}}_{\delta(t-t')}$$

$$\Rightarrow G(\omega) = \frac{-1}{-\omega^2+\Omega^2} = \frac{1}{\omega^2-\Omega^2} \quad \text{(easy)}$$

$$\Rightarrow G(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{\omega^2-\Omega^2}$$

But the integral has two poles, at $\pm\Omega$, on the real axis $\Rightarrow$ how to integrate about them?

<u>Prescription</u> $\omega^2 \to \omega^2 + i\varepsilon$, $\varepsilon = +0$

which leads, as we'll see, to <u>time-ordered</u> <u>GF</u>

There are two other prescriptions — retarded and advanced — which are also used frequently.

But time-ordered one is particularly suited for path integrals (as shown by Feynman and many others after him).

$$S_{cl} = \int_0^T dt \left(\tfrac{1}{2} m \dot{x}_{cl}^2 - \tfrac{1}{2} m \Omega^2 x_{cl}^2 + J x_{cl} \right)$$

$$\int dx\, \dot{x}_{cl}^2 = \int dx\, \dot{x}_{cl} \dot{x}_{cl} = \int_0^T dx \left(\frac{d}{dt} (\dot{x}_{cl} x_{cl}) - \ddot{x}_{cl} x_{cl} \right)$$

$$= \underbrace{\dot{x}_{cl} x_{cl} \Big|_0^T}_{\text{can not say it is } \emptyset.} - \int_0^T dt \underbrace{\left(x_{cl} \frac{d^2}{dt^2} x_{cl} \right)}_{x_{cl} \ddot{x}_{cl}}$$

But $x = x_{cl} + y \Rightarrow S = S_{cl} + S^{(2)}$ where

$$S^{(2)} = \int_0^T dt \left(\tfrac{1}{2} m \dot{y}^2 - \tfrac{1}{2} m \Omega^2 y^2 \right) \quad (\text{no } J(t) y \text{ term!})$$

and now $y(0) = y(T) = 0$

$$\Rightarrow S^{(2)} = \int_0^T dt\; y \left(-\tfrac{1}{2} m \frac{d^2}{dt^2} - \tfrac{1}{2} m \Omega^2 \right) y =$$

Next, $F = D(0,T;0,0) = \int Dy\; e^{\frac{i}{\hbar} S^{(2)}(y)} =$

$$= \int Dy\; \exp\left\{ -\frac{m}{2} \frac{i}{\hbar} \int_0^T dt\, y \underbrace{\left(\frac{d^2}{dt^2} + \Omega^2 \right)}_{\text{this is the same operator as in } \left(\frac{d^2}{dt^2} + \Omega^2\right) G(t) = -\delta(t)} y \right\}$$

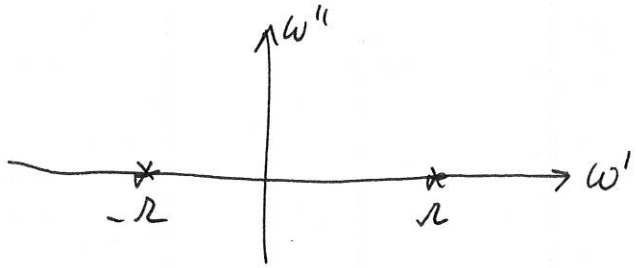
For F to exist we need a convergence factor $\left(\frac{d^2}{dt^2} + \Omega^2 - i\varepsilon \right)$, with $\varepsilon = +0$, so that $\exp\left\{ -\frac{m\varepsilon}{2\hbar} \int_0^T dt\, y^2(t) \right\}$ is present and makes everything well-behaving.

$$G(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} G(\omega)$$

$$\left(\frac{d^2}{dt^2} + \Omega^2\right) G(t) = -\delta(t) = -\int \frac{d\omega}{2\pi} e^{-i\omega t}$$

$$\int \frac{d\omega}{2\pi} \left\{ \left(-\omega^2 + \Omega^2\right) G(\omega) = -1 \right\}$$

$$G(\omega) = \frac{1}{\omega^2 - \Omega^2}$$ has poles at $\omega = \pm\Omega$ on real axis of ω



Time-ordered GF

Ω = ~~[illegible]~~ resonance frequency of the oscillator

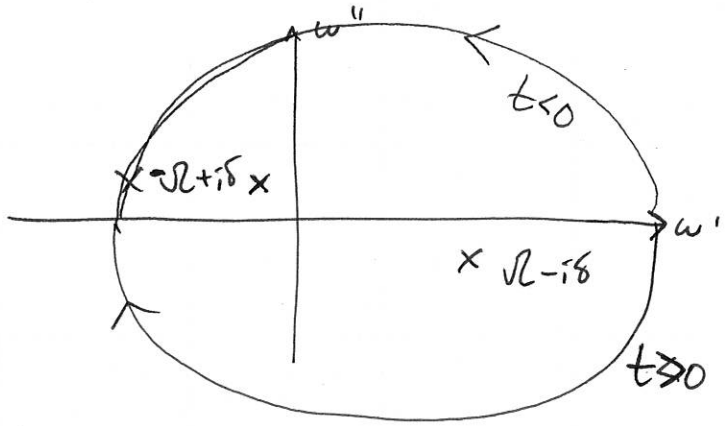
$$G(\omega) = \frac{1}{\omega^2 - \Omega^2 + i\varepsilon} \qquad (\varepsilon = +0)$$

$$\omega^2 = \Omega^2 - i\varepsilon \qquad \omega = \pm\sqrt{\Omega^2 - i\varepsilon} = \pm\Omega\left(1 - \frac{i\varepsilon}{2\Omega^2}\right)^{1/2}$$

$$= \pm\Omega\left(1 - \frac{i\varepsilon}{2\Omega^2}\right) = \begin{cases} \Omega - \frac{i\varepsilon}{2\Omega} \\ -\Omega + \frac{i\varepsilon}{2\Omega} \end{cases}$$

$$G(\omega) = \frac{1}{(\omega - \Omega + i\delta)(\omega + \Omega - i\delta)} \qquad \delta \equiv \frac{\varepsilon}{2\Omega} = +0$$

$$G(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{(\omega - \Omega + i\delta)(\omega + \Omega - i\delta)}$$



$$e^{-i(\omega'+i\omega'')t} = e^{-i\omega' t + \omega'' t}$$

1) let $t>0 \Rightarrow$ need to integrate over the region where $\omega''<0$ i.e. lower half-plane

$$\Rightarrow G(t) = \frac{-2\pi i}{2\pi} \frac{e^{-it(\Omega - i\delta)}}{2(\Omega - i\delta)} = \frac{e^{-i\Omega t} e^{-\delta t}}{2i\Omega} \quad (t>0)$$

2) $t<0 \Rightarrow$ need $\omega''>0 \Rightarrow$ upper half-plane

$$G(t) = \frac{2\pi i}{2\pi} \frac{e^{-it(-\Omega + i\delta)}}{-2(\Omega - i\delta)} = \frac{e^{it\Omega} e^{\delta t}}{2i\Omega} \quad (t<0)$$

$$G(t) = \theta(t) \frac{e^{-i\Omega t}}{2i\Omega} + \theta(-t) \frac{e^{i\Omega t}}{2i\Omega}$$

Note that for $|t| \to \infty$ $G(t) \to 0$ due to $e^{-\delta|t|}$ convergence factor

Time-ordered correlation functions

$$D(x', T; x, 0) = \int Dx\, e^{\frac{i}{\hbar} S[x]} = \langle x', T | x, 0 \rangle$$

We are often interested in finding out matrix elements of various operators: $\hat{X}(t)$ = operator in Heisenberg representation

$$\langle x', T | \hat{X}(t_1) \hat{X}(t_2) | x, 0 \rangle \qquad \underline{(\text{let } T > t_1 > t_2 > 0)}$$

$$= \int dx_1 dx_2 \; \langle x', T | \hat{X}(t_1) \underbrace{| x_1, t_1 \rangle \langle x_1, t_1 |}_{\int dx_1 |x_1\rangle\langle x_1| = 1} \hat{X}(t_2) \underbrace{| x_2, t_2 \rangle \langle x_2, t_2 |}_{\int dx_2 |x_2\rangle\langle x_2| = 1} x, 0 \rangle$$

$$= \int dx_1 dx_2 \; x_1 x_2 \underbrace{\langle x', T | x_1, t_1 \rangle}_{D(x', T; x_1, t_1)} \underbrace{\langle x_1, t_1 | x_2, t_2 \rangle}_{D(x_1, t_1; x_2, t_2)} \underbrace{\langle x_2, t_2 | x, 0 \rangle}_{D(x_2, t_2; x, 0)}$$

integrating over all $x_{1,2}$!

$$D(x', T; x_1, t_1) = \int Dx\, e^{\frac{i}{\hbar} \int_{t_1}^{T} dt' L}$$

$$D(x_1, t_1; x_2, t_2) = \int Dx\, e^{\frac{i}{\hbar} \int_{t_2}^{t_1} dt' L}$$

$$D(x_2, t_2; x, 0) = \int Dx\, e^{\frac{i}{\hbar} \int_{0}^{t_2} dt' L}$$

$$= \int Dx \; \cancel{x(t_1) x(t_2)}\; x(t_1) x(t_2)\, e^{\frac{i}{\hbar} \int_0^T dt' L}$$

since $\int_{t_1}^{T} dt' \ldots + \int_{t_2}^{t_1} dt' \ldots + \int_0^{t_2} dt' \ldots = \int_0^T dt' \ldots$

What if $\underline{t_2 > t_1}$:

$$\langle x', T | \hat{X}(t_2) \hat{X}(t_1) | x, 0 \rangle = \int dx_2 dx_1 \langle x', T | \hat{X}(t_2) | x_2, t_2 \rangle \cdot$$

$$\cdot \langle x_2, t_2 | \hat{X}(t_1) | x_1, t_1 \rangle \langle x_1, t_1 | x, 0 \rangle =$$

$$= \int dx_1 dx_2 \; x_2 x_1 \; D(x', T; x_2, t_2) D(x_2, t_2; x_1, t_1)$$

$$D(x_1, t_1; x, 0) = \int Dx \; \underbrace{x(t_2) x(t_1)}_{x(t_1) x(t_2)} \; e^{\frac{i}{\hbar} \int_0^T dt' L}$$

because these are classical

we actually are dealing here with

$\Rightarrow$ time-ordered matrix elements

$$\langle x', T | \hat{T}(\hat{x}(t_1)\hat{x}(t_2)) | x, 0\rangle = \int Dx \; x(t_1)x(t_2)\, e^{\frac{i}{\hbar}S[x]}$$

$\hat{T}$ = operator of time-ordering (chronological ordering)

$$\hat{T}(\hat{x}(t_1)\hat{x}(t_2)) = \theta(t_1-t_2)\,\hat{x}(t_1)\hat{x}(t_2) + \theta(t_2-t_1)\,\hat{x}(t_2)\hat{x}(t_1)$$

This clearly is easily generalized to any number of operators

$$\langle x', T | \hat{T}(\hat{O}_\alpha(t_1)\,\hat{O}_\beta(t_2)\ldots\hat{O}_\gamma(t_n)) | x, 0\rangle =$$

$$= \int Dx \; \hat{O}_\alpha(x(t_1))\,\hat{O}_\beta(x(t_2))\ldots\hat{O}_\gamma(x(t_n))\, e^{\frac{i}{\hbar}S[x]}$$

~~We often are interested in transition~~

Let's now look on a system with a <u>source term</u> added to the action: $L \to L + Jx$

$$D_J(x', T; x, 0) = \int Dx \; e^{\frac{i}{\hbar}\int_0^T dt\,[L + J(t)x(t)]}$$

Observe that $(T > t_1 > 0)$

$$\frac{\delta}{\delta J(t_1)} e^{\frac{i}{\hbar}S_J[x]} = \frac{\delta}{\delta J(t_1)} e^{\frac{i}{\hbar}\int_0^T dt\,(L + J(t)x(t))} =$$

$$= \frac{i}{\hbar}\frac{\delta S_J[x]}{\delta J(t_1)} e^{\frac{i}{\hbar}S_J[x]} = \frac{i}{\hbar}\,x(t_1)\,e^{\frac{i}{\hbar}S_J[x]}$$

As a result, $\frac{\delta}{\delta J(t_1)} D_J(x', T; x, 0) =$

$$= \frac{\delta}{\delta J(t_1)} \langle x', T | x, 0 \rangle_J = \langle x', T | \hat{x}(t_1) | x, 0 \rangle_J \left(\frac{i}{\hbar}\right)$$

Clearly, $\frac{\delta^2}{\delta J(t_1)\, \delta J(t_2)} \langle x', T | x, 0 \rangle_J = \left(\frac{i}{\hbar}\right)^2 \langle x', T | \cdot ($

$(T > t_1, t_2 > 0)$

$$\left(\hat{T}\{ \hat{x}(t_1)\, \hat{x}(t_2) \} \right) | x, 0 \rangle_J$$

$\Rightarrow$ We can find time-ordered functions by first calculating transition amplitude $D_J(x', T; x, 0)$ (in the presence of external sources) and then taking functional derivatives

$$\frac{\delta}{\delta J(t_j)} .$$

Most often, we are interested in transition amplitude between different <u>states</u> $|\psi_i\rangle$ at time t_i, $|\psi_f\rangle$ at $t_f > t_i$

$$\langle \psi_f | \psi_i \rangle = \int dx_f dx_i \langle \psi_f | x_f, t_f \rangle \langle x_f, t_f | x_i, t_i \rangle \underbrace{\langle x_i, t_i | \psi_i \rangle}_{\psi_i(x_i, t_i)}$$

$$= \int dx_f \, dx_i \, \psi_f^*(x_f, t_f) \, \psi_i(x_i, t_i) \cdot$$

$$\cdot \int Dx \, e^{\frac{i}{\hbar} S[x]}$$

Hence $\langle \psi_f | \hat{T}\{\hat{O}(t_1)\hat{O}(t_2)\ldots\hat{O}(t_n)\} | \psi_i \rangle =$

$$= \int dx_f \, dx_i \, \psi^*(x_f, t_f) \, \psi_i(x_i, t_i) \cdot$$

$$\cdot \int Dx \, O(x(t_1)) \, O(x(t_2)) \ldots O(x(t_n)) \, e^{\frac{i}{\hbar} S[x]}$$

In the case of <u>expectation value</u> the matrix element is $\langle \psi_i | \hat{T}\{O(t_1)\ldots O(t_n)\} | \psi_i \rangle =$

$$= \int dx_f \, dx_i \, \psi_i^*(x_f, t_f) \, \psi_i(x_i, t_i) \cdot$$

$$\cdot \int Dx \, O(x(t_1)) \ldots O(x(t_n)) \, e^{\frac{i}{\hbar} S[x]}$$

The state $|\psi_i\rangle$ is not necessarily normalized
$\Rightarrow$ need to divide by its norm $\langle \psi_i | \psi_i \rangle$

$$\Rightarrow \langle \psi_i | \hat{T}\{\hat{O}(t_1)\ldots\hat{O}(t_n)\} | \psi_i \rangle =$$

$$= \frac{\int dx_f \, dx_i \, \psi_i^*(x_f, t_f) \, \psi_i(x_i, t_i) \int Dx \, O(x(t_1)) \ldots O(x(t_n)) \, e^{\frac{i}{\hbar} S[x]}}{\int dx_f \, dx_i \, \psi_i^*(x_f, t_f) \, \psi_i(x_i, t_i) \int Dx \, e^{\frac{i}{\hbar} S[x]}}$$

Note that this expression does not involve any additional normalization factors.